

Hands on Homework #4 (Equivalent Systems 3D)

EGN 3310 - 0M01 Engineering Analysis: STATICS



MWF, 11:30AM-12:20PM

Submitted by: Group 38

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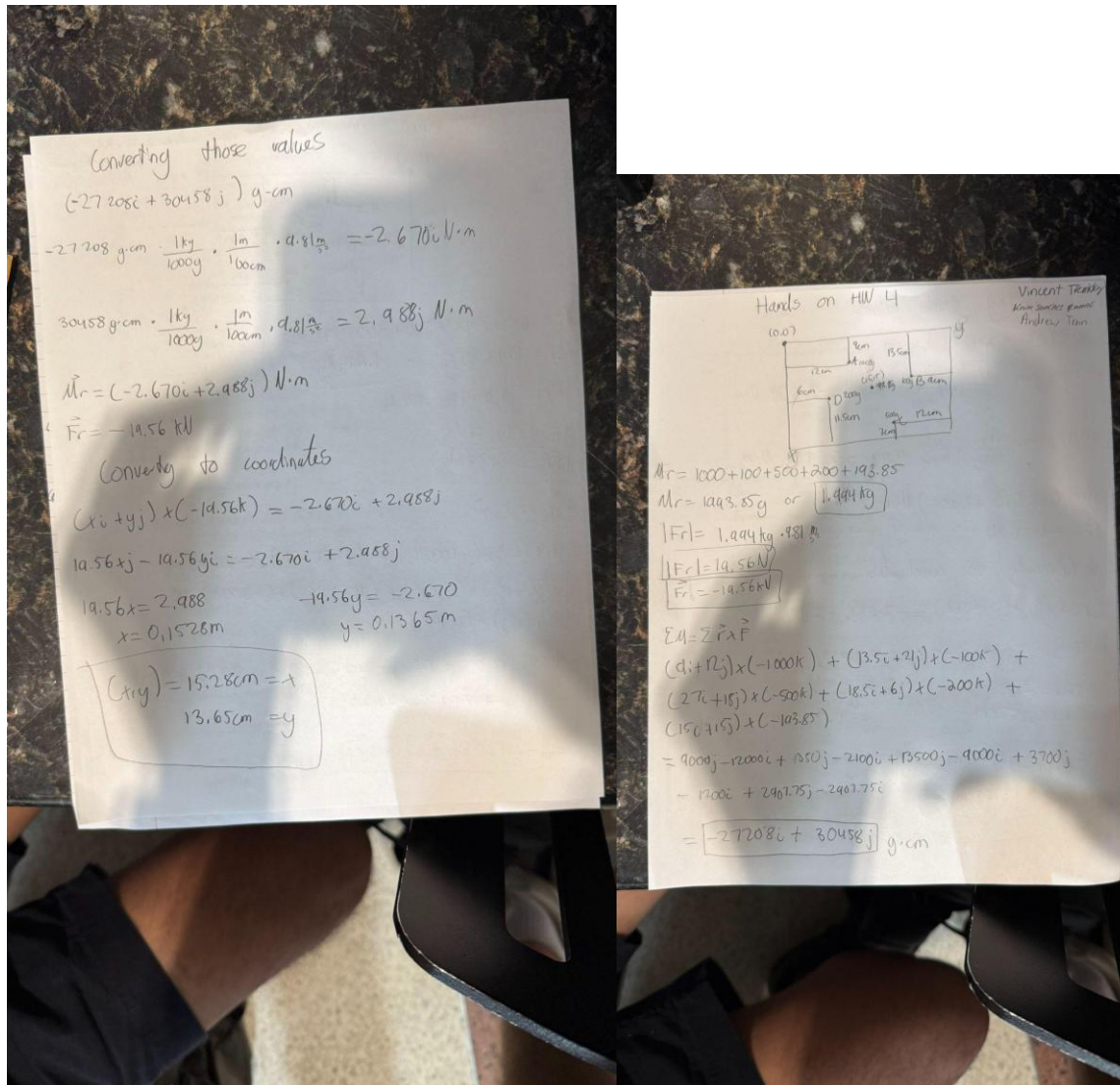
Kevin Sanchez Ramos - ke987766

Hand Calculation (Theoretical Method)

Pictures of our board on scale:



Here is our paper will all our calculations:



Starting off we picked random points for each of our masses and set the mass of the board acting at the middle of our 30x30cm board, which is 15,15. The paper has that information, but here is a table to help you as well.

	X	Y	Mass(g)
A	9	12	1000

B	13.5	21	100
C	27	18	500
D	18.5	6	200
Board	15	15	193.58

Following this our first step was to find the resultant force, which is done by adding each of the masses together, converting grams to meters, and then multiplying them by 9.81.

$$(1000 + 500 + 200 + 100 + 193.58)g * (1m/1000g) * 9.81m/s^2$$

Therefore our $F_r = -19.56k \text{ N}$

****Note:** k in this case is the direction of the vector, not “kilo”-Newtons

Following this, we found the summation of moments. This had to be done using the cross product of the z axis (the masses) and the points on the x-y axis.

$$\begin{aligned} \sum M = & (9i + 12j) \times (-1000k) + (13.5i + 21j) \times (-100k) + (27i \\ & + 18j) \times (-500k) + \\ & (18.5i + 6j) \times (-200k) + (15i + 15j) \times (-193.88k) \end{aligned}$$

This required multiple cross products, and then we added them all up in terms of i and j at the end. This gave us $-27208i + 30458j \text{ g*cm}$.

In order to convert these into the final coordinates, we had to convert these g*cm into N*m so it is compatible with the F_r we found above.

$$(-27208i + 30458j) * \frac{1kg}{1000g} * \frac{1m}{100cm} * 9.81 \frac{m}{s^2} = (-2.670i + 2.988j) \text{ N * m}$$

This now meant that we had both the vector of the resultant moment, as well as that of the resultant force. Here they are listed once again

$$\vec{Mr} = (-2.670i + 2.988j) \text{ N * m}$$

$$\vec{Fr} = -19.56k \text{ N}$$

****Note:** k in this case is the direction of the vector, not “kilo”-Newtons
Now that we have everything we need, we go back to the formula for momentum to find the coordinate of the resultant force.

$$M = r \times F$$

$$(xi + yj) \times (-19.56k) = (-2.670i + 2.988j)$$

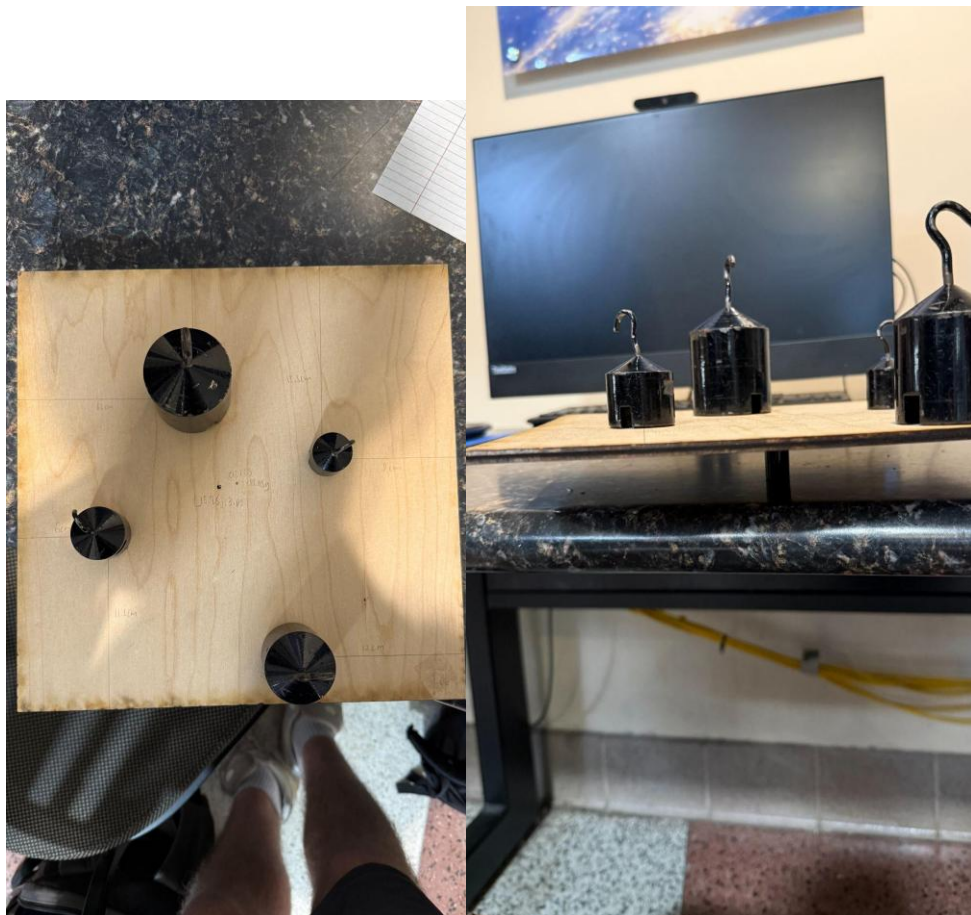
Solving for x and y we get

$$x = 0.1528m \text{ or } 15.28cm$$

$$y = 0.1365m \text{ or } 13.65 \text{ cm}$$

That gives us the locations of the resultant force, which concludes the calculations portion. We will now apply that to the board to see if our calculations were correct.

Hands-On Method (Physical Model)



As you can see above, we managed to balance each of the weights on the cap at the point we calculated in the previous section (15.28,13.65). This means that the resultant force acts through that point, so the resultant force is at (15.28,13.65). For each of the masses, we allocated them to a letter, and marked their coordinates from the origin. Below is a table with the values for each point.

Table of information for each mass (refer to the image to see which is allocated to what letter)

	X	Y	Mass(g)
A	9	12	1000
B	13.5	21	100
C	27	18	500
D	18.5	6	200
Board	15	15	193.58

With the information, this meant we had no percent error for this, since the point we calculated and the point we actually balanced it at is the exact same. However this doesn't mean we have no sources of error.

One potential source is the fact that the masses will not be concentrated, even if we assume so at a specific point. They will provide a distributed load on the board, so it's not possible to fully assume that.

Furthermore, small bends on the board and the smoothness of it also affect the results as well. Rounding errors can cause small shifts in the distance, which can be critical especially with such a small board.

Other notable errors could be the placement of the masses on the board, which could've been off by a few millimeters of the original measurements. Lastly, we also drilled a small hole through the center of gravity to help with the placement of the balancing point (which we used a marker cap for), which could've slightly altered the real center of gravity since there's less mass and less area making contact with the cap.

This concludes the hands-on section. This was a simpler one, as we did the calculations beforehand and simply applied them to the board. As for us, we did great since our calculations directly applied to the board, which is great!

Conclusion/discussion:

Vincent Tremblay (vi474918):

Overall I really liked the way this lab was made. It was a great opportunity to see how calculations can transfer to real world use, which we didn't get to see much in the previous labs. In our case, we did really well as the coordinates we calculated were the right ones to balance the board. Of course, there were some sources of error as we previously mentioned, but ultimately we did a great job and were able to apply the concepts of moments and cross product to this.

Andrew Tran (an365172):

I believe that this hands-on assignment up to this point has not only been the most successful hands-on assignment but also one of the most important ones we've done so far. Even with the potential sources of errors, we were able to find the center of gravity and make the wood platform stable. It is evidence that you can calculate center of gravity and make a platform stable just by calculating resultant force and moment with respect to a point. As a result, this hands-on assignment also has a lot of real world applications, so it was very satisfying to learn, apply, and then construct a model to demonstrate how engineers and computers use moments and forces to get the center of gravity (the area there is 0 moments - no torque).

Kevin Sanchez Ramos (ke987766):

From this experiment, using our understanding of statistics and the resources available to us, I believe this was a really good opportunity to learn and apply real-world applications as a student. From our measurements and calculations, I believe we did well in understanding the concept behind what we are learning, while also recognizing other potential factors that could affect our experiment, such as the fact that the forces were based on a single point rather than the distributed forces our weights showed. Ultimately, this hands-on homework was fascinating, knowing that we can conduct experiments like this in the future as engineers.